



PHYSICS HSSC-I SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ سیاہ پینسل کا استعمال ممنوع ہے۔

Version No.				
3	2	0	4	1

ROLL NUMBER						

- ○ ● ○ ○
① ① ① ① ●
② ● ② ② ②
● ③ ③ ③ ③
④ ④ ④ ● ④
⑤ ⑤ ⑤ ⑤ ⑤
⑥ ⑥ ⑥ ⑥ ⑥
⑦ ⑦ ⑦ ⑦ ⑦
⑧ ⑧ ⑧ ⑧ ⑧
⑨ ⑨ ⑨ ⑨ ⑨

- ○ ○ ○ ○ ○ ○
① ① ① ① ① ① ①
② ② ② ② ② ② ②
③ ③ ③ ③ ③ ③ ③
④ ④ ④ ④ ④ ④ ④
⑤ ⑤ ⑤ ⑤ ⑤ ⑤ ⑤
⑥ ⑥ ⑥ ⑥ ⑥ ⑥ ⑥
⑦ ⑦ ⑦ ⑦ ⑦ ⑦ ⑦
⑧ ⑧ ⑧ ⑧ ⑧ ⑧ ⑧
⑨ ⑨ ⑨ ⑨ ⑨ ⑨ ⑨

Answer Sheet No. _____

Invigilator Sign. _____ ہر سوال کے سامنے دیے گئے، کرکولم کے مطابق درست دائرہ کو پر کریں۔

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. In a simple pendulum experiment, percentage errors (uncertainty) in "L" and "T" are 0.1% and 2% respectively. Maximum percentage error (uncertainty) in "LT ² " is:	4.1%	3%	2.1%	3.2	○	○	○	○
2. If $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B} $ then angle "θ" between vectors $\vec{A}$ and $\vec{B}$ is:	0	$\pi/4$	$\pi/2$	π	○	○	○	○
3. When a massive body of mass M ₁ is moving with velocity V ₁ collides with lighter stationary body of mass m ₂ , the velocity of massive body after collision will be:	$V_1' = 2V_1$	$V_2' = V_1$	$V_1' = V_1$	$V_2' = 2V_2$	○	○	○	○
4. Moment of inertia of a body spinning about its axis DOES NOT depend on the:	Angular velocity of body	Mass of body	Orientation of the axis	Distribution of mass around axis	○	○	○	○
5. The bodies A and B of equal masses have velocities ratio in 2:1 their K.E are in the ratio of:	2:1	1:2	4:1	1:4	○	○	○	○
6. The fluid is said to be incompressible, if its density is:	Constant	Very high	Very small	Zero	○	○	○	○
7. Equation of continuity is based upon law of conservation of:	Momentum	Energy	Mass	Charge	○	○	○	○
8. Glass is an example of _____ solid.	Crystalline	Amorphous	Poly-crystalline	Ductile	○	○	○	○
9. Young's Modulus, Y=	$\frac{\Delta L / L}{F / A}$	$\frac{FA}{\Delta L L}$	$\frac{\Delta L L}{FA}$	$\frac{F / A}{\Delta L / L}$	○	○	○	○
10. A Carnot engine undergoes _____ processes in one cycle.	Two	Three	Four	Six	○	○	○	○
11. Stars moving away from the earth show:	Red shift	Blue shift	Black shift	Yellow shift	○	○	○	○
12. A negatively charged particle is placed in a uniform electric field directed from south to north. In which direction will the particle move after being released?	North	South	East	West	○	○	○	○
13. A 100 ohm resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	1 watt	10 watts	100 watts	1000 watts	○	○	○	○
14. The SI unit of magnetic induction is:	Nm	Gauss	Weber	Tesla	○	○	○	○
15. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero	○	○	○	○
16. Inertial frame of reference is a frame in which _____ law of motion holds.	1 st	2 nd	3 rd	4 th	○	○	○	○
17. Two up quarks and one down quark make a:	Meson	Photon	Neutron	Proton	○	○	○	○

— 1HA-I 25004 (B) —

• $(K.E)_{max} = \frac{1}{2}mv^2$ • $l = mr^2$ • $P = VI$ • $P = \frac{V^2}{R}$ • $\vec{A} \cdot \vec{B} = AB \cos \theta$ • $\vec{A} \times \vec{B} = AB \sin \theta$ • $L = I\omega$ • $v = r\omega$



PHYSICS HSSC-I

22

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly.

(14 x 3 = 42)

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

(i)	Show that the equation $P = \rho gh$ is dimensionally homogeneous.	03	OR	Explain that increase in entropy means degradation of energy.	03
(ii)	Time for 10 vibrations of a simple pendulum is 17.6s. Least count of stop watch is 0.1s. Find time period of simple pendulum. Also determine uncertainty in it.	03	OR	When water falls from a tap, its cross-sectional area decreases as it comes down. Explain why? 	03
(iii)	Under what condition the rectangular (perpendicular) components of a vector have same magnitude?	03	OR	Discuss the motion of two bodies after collision when $m_1 \ll m_2$ and $V_2 = 0$.	03
(iv)	What is meant by Gyroscope? Explain briefly.	03	OR	What is meant by work? How can work done be calculated graphically by a variable force?	03
(v)	A space ship has diameter of 8m rotates about its axis to produce artificial gravity of magnitude 17.47ms^{-2} in it. Calculate with which angular speed the satellite rotates?	03	OR	Differentiate between conservative and non-conservative forces. Give examples.	03
(vi)	Derive the relation between torque, moment of inertia and angular acceleration.	03	OR	Derive the formula for Kinetic Energy [using the equations of motion]	03
(vii)	What is meant by Magnus effect? Give its causes and effects.	03	OR	Differentiate between reversible and irreversible processes. Also give examples.	03
(viii)	Show that: $\frac{\text{Volt}}{\text{meter}} = \frac{\text{Newton}}{\text{Coulomb}}$	03	OR	Find the power delivered by an engine for attaining velocity $= (4\hat{i} + 5\hat{j})$ m/s while it exerts a force $= (8\hat{i} - 2\hat{j})$ N.	03
(ix)	What is meant by null method of measurement? Enlist two applications.	03	OR	Briefly describe the terms, stress, strain and Young's modulus.	03
(x)	Why are the suspension bridges written with their period of use? Why after that period it is dangerous to use the bridge? (Elastic potential)	03	OR	What is meant by magnetic flux linkage? Explain briefly.	03
(xi)	State three statements of second law of thermodynamics.(one in terms of entropy)	03	OR	How is Doppler's effect helpful in diagnosis of cardiac diseases? Explain briefly.	03
(xii)	Calculate entropy change when 800g of ice melts at 0°C , latent heat of fusion for ice is $3.36 \times 10^5 \text{ J/kg}$. Also discuss sign of entropy.	03	OR	Can sound waves be polarized? Explain briefly.	03
(xiii)	To increase intensity of waves by a factor of 25, calculate the required increase in amplitude of wave.	03	OR	What is super fluidity? Give some of its applications.	2+1
(xiv)	Under what conditions can the terminal potential differences of a battery exceed its emf?	03	OR	State the characteristics of vector product.	03

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	What is meant by projectile? Determine the expressions for velocity and displacement of a projectile thrown horizontally from a certain height.	1+4 +2	OR	What is meant by electric field strength? Show that: $E = \frac{-\Delta V}{\Delta r}$	07
Q.4	State and explain terminal velocity. Derive its equation by using stoke's law.	1+2 +3	OR	What is meant by quark? Explain the quark family of particles.	2+4
Q.5	What are stationary waves? Discuss the formation of harmonics in stationary waves in an open organ pipe.	1+6	OR	What is centripetal acceleration? Derive an expression for centripetal acceleration in terms of angular velocity " ω ".	1+6
Q.6	Discuss and derive an expression for magnetic force acting per unit length of conductor, when current is passing through two parallel straight conductors in upward direction.	06	OR	Discuss the following consequences of special theory of relativity: i) Time dilation ii) Length contraction iii) Mass increase	2x3

— 1HA-I 25004 (D) —

$$\begin{aligned}
 & \Delta S = \frac{\Delta Q}{T} & v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) v_1 & T = \frac{\text{Total time}}{\text{No. of vibrations}} & P = \bar{F} \cdot \bar{v} & I \propto A^2 & \omega = \sqrt{\frac{g}{R}} & \text{uncertainty} = \frac{\text{Absolute uncertainty}}{\text{No. of vibrations}} \\
 & \Delta Q = mL_f & v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_1 & W = KE = F \cdot \Delta d & F_{\text{drag}} = 6\pi\eta r v & P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2
 \end{aligned}$$